1. (a) Definitions:

   - $\Omega \neq \emptyset$, $\mathcal{C} \subset \mathcal{P}(\Omega)$ is semi-algebra if
     (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$($\pi$-system);
     (ii) $\forall A \in \mathcal{C} \Rightarrow A^C = \bigcup_{i=1}^k B_i, \{B_i\}_{i=1}^k \subset \mathcal{C}$ are disjoint;

   - $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an algebra if
     (i) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$;
     (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
     (iii) $\Omega \in \mathcal{F}$.

   - $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an σ-algebra if
     (i) $\{A_i\} \subset \mathcal{F} \Rightarrow \bigcup_{i=1}^\infty A_i \in \mathcal{F}$;
     (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
     (iii) $\Omega \in \mathcal{F}$.

   (b) ("only if" part) If $\mathcal{A}$ is an algebra, then $\Omega \in \mathcal{A}$ and $A \setminus B = A \cap B^c \in \mathcal{A}$ since algebras are closed under complements and intersections. For the "if part", suppose $\mathcal{A}$ satisfies the two properties. Only need to verify that $\mathcal{A}$ is closed under complements and intersections (or equivalently, unions): $A \in \mathcal{A}$ then $A^c = \Omega \setminus A \in \mathcal{A}$ from the two properties of $\mathcal{A}$. Also if $A, B \in \mathcal{A}$, then $B^c \in \mathcal{A}$ (already verified) and so $A \cap B = A \setminus B^c \in \mathcal{A}$. This completes the proof.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

   If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

   If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\cup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\cup_{i \geq 1} A_i = (\cup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ-algebra of subsets of Ω_0.

3. (a) $\Omega = \mathbb{N}$, $\mathcal{F} = \mathcal{P}(\mathbb{N})$, $\mu(A_n) = \infty$, $\bigcap_{n \geq 1} A_n = \emptyset$. So $\mu(\bigcap_{n \geq 1} A_n) \neq \lim_{n \to \infty} \mu(A_n)$. Hence m.c.f.a doesn't hold.

   (b) Take $\mu'(A) = \sum_{i \in A} \frac{1}{2^i}$ (or we can choose any discrete probability measure). Then $\mu(\bigcap_{n \geq 1} A_n) = 0 = \lim_{n \to \infty} \frac{1}{2^{n-1}} = \lim_{n \to \infty} \mu(A_n)$. So m.c.f.a holds.

   (c) If $\exists\ n_0$ s.t $\mu(A_{n_0}) < \infty$, then m.c.f.a holds.

   (d) Always true.

4. (a) Let $\mathcal{C} = \{(a,b] : -\infty \le a < b < \infty\} \cup \{(a,\infty), a \in \mathbb{R}\}$. It is a semi-algebra.
Define

$$\begin{aligned}
\mu_F(a,b] &= F(b+) - F(a+) \\
&= (1-e^{-b}) - (1-e^{-a}) \\
&= e^{-a} - e^{-b} \\
\mu_F(a,\infty] &= F(\infty-) - F(a+) \\
&= 1 - (1-e^{-a}) \\
&= e^{-a}
\end{aligned}$$

Then this defines a measure on semi-algebra $\mathcal{C}$.
By Caratheodory Extension Theorem, there exists an extension of μ_F, called μ_F^* on $(\Omega, \mathcal{M}_{\mu_F^*}, \mu_F^*)$, where $\mathcal{M}_{\mu_F^*}$ = set of all μ_F^*-measurable sets on $\Omega = \{A \in \mathcal{P}(\Omega) : \mu_F^*(E) = \mu_F^*(E \cap A) + \mu_F^*(E \cap A^C), \forall E \in \Omega\}$. $\mathcal{M}_{\mu_F^*}$ is a σ-algebra containing $\mathcal{C} \Rightarrow \mathcal{B}(\mathbb{R}) \subset \sigma\langle\mathcal{C}\rangle \subset \mathcal{M}_{\mu_F^*}$. μ_F^* restricted to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is an extension of μ_F.

(b) This is unique since $\mu_F^*(\mathbb{R}) = 1$ (finite measure, hence any other measure that takes same value on $\mathcal{C}$ will take same values on $\mathcal{B}(\mathbb{R})$).